MODULE 07 · DOMAIN 1 - DATA PREPARATION FOR ML AND AI

Labelling and human review

Ground Truth, Mechanical Turk and A2I. Examinable, and closed to new customers.

MODULE 07 of 43 DOMAIN 1 - Data Prep WEIGHT 28%

admin@sthithapragna.com

What is actually true about these services

- **Ground Truth, Mechanical Turk and A2I:** closed to new customers, 30 July 2026.
- **Ground Truth Plus:** end of support reached 30 June 2026. It is gone.
- Existing customers continue as normal. Security and availability work continues.
- No new features are planned for any of them.
- **All are named in the exam guide, so all are examinable.** Answer as the guide expects.

> ON THE EXAM
>
> If you open a new account to practise and cannot create a labelling job, nothing is broken. That is the expected behaviour now.

For supervised learning, **labels are the constraint**. Not compute, not algorithms.

Why labelling dominates project timelines

- Cost scales with **examples times seconds per example times reviewers**.
- Segmentation masks take tens of seconds each. Hundreds of images become hours.
- Specialist labelling, such as clinical or legal, needs experts and costs far more.
- Every strategy here attacks one of those three multipliers. That is the whole design space.
- Getting labels wrong is worse than having fewer. Noise propagates into every prediction.

> ON THE EXAM
>
> AWS positions data labelling as reducing costs by up to 40 percent. The lever is always fewer human seconds per example.

From raw objects to a labelled manifest

01

Input

A manifest listing objects in S3, one per line.

02

Task

A task type and a worker UI template define the job.

03

Workforce

Private, Mechanical Turk, or an AWS Marketplace vendor.

04

Output

An augmented manifest: original object plus its label.

ON THE EXAM

The augmented manifest is the useful output. It feeds straight into training, including Pipe mode from module 02.

Input manifest in, augmented manifest out

FILE	WHAT IT CONTAINS
Input manifest	JSON Lines. One object per line, usually an S3 URI as source-ref
Output manifest	The same lines, plus the label and label metadata attached
Label metadata	Confidence, which worker or model produced it, the job name, a timestamp
Why it matters	Training can read the augmented manifest directly, including in Pipe mode

ON THE EXAM

The metadata records whether a label came from a human or from automated labelling. Keep it; it explains the dataset later.

Three workforces, and the rule that separates them

WORKFORCE	WHEN IT IS RIGHT
Private	Sensitive data, or domain expertise. Your own employees or contractors
Amazon Mechanical Turk	500,000+ workers, 24/7, cheapest. Non-sensitive public data only
Vendor (AWS Marketplace)	Specialist skills at scale. Prescreened by AWS for security

ON THE EXAM

Sensitivity decides this, not cost. Confidential, personal or regulated data rules out Mechanical Turk outright.

How a private workforce is actually created

- Backed by an **Amazon Cognito user pool**, created for you or supplied by you.
- Or by your own **OIDC identity provider**, so labellers use existing corporate logins.
- Workers sign in to a **worker portal** with its own URL, separate from the AWS console.
- **Work teams** subdivide a workforce, so different groups get different task types.
- One private workforce per Region per account. Work teams are how you segment it.

> ON THE EXAM
>
> Labellers never need AWS console access. They log in to the worker portal, which is the point of the separation.

Built-in task types

- **Images:** classification, bounding boxes, semantic segmentation, label verification.
- **Text:** classification, named entity recognition.
- **Video:** clip classification, object detection and object tracking across frames.
- **3D point clouds:** object detection and tracking, for lidar data.
- **Custom task types** use your own HTML template for anything not covered.

> ON THE EXAM
>
> Label verification is the one people overlook: humans reviewing existing labels rather than creating new ones.

The two Lambda functions in a custom labelling job

FUNCTION	WHAT IT DOES
Pre-annotation Lambda	Runs before a worker sees the object. Shapes the data into the template
Post-annotation Lambda	Runs after workers finish. Consolidates several answers into one label
Worker task template	Crowd HTML elements defining the interface the worker uses
Together	These three are what make a custom task type possible at all

ON THE EXAM

Annotation consolidation is not magic for custom types. The post-annotation Lambda is where you write that logic yourself.

Automated data labelling, or active learning

- Humans label a sample. A model trains on it and labels the rest automatically.
- Only objects where the model is **uncertain** go back to humans. The rest are auto-labelled.
- You set a confidence threshold, so you control the accuracy and cost trade-off.
- This is the single biggest cost reduction available on a large labelling job.
- It needs enough volume to be worth it. On a small dataset it is overhead.

> ON THE EXAM
>
> The pattern is: let the model do the easy examples and spend human attention only where the model hesitates.

Getting labels you can trust

- **Annotation consolidation:** several workers label the same object, results are merged.
- Disagreement between workers is a **signal**. It usually means ambiguous instructions.
- Clear instructions with worked examples do more for quality than more reviewers do.
- More workers per object multiplies cost linearly. Use it where ambiguity is real.
- Run a small pilot batch and inspect it before committing to the full dataset.

> ON THE EXAM
>
> If three workers disagree about the same image, the task definition is usually the problem, not the workers.

Streaming labeling jobs, and their limits

- A long-running job that accepts **new objects while it is active**.
- **API only.** You cannot create a streaming labelling job in the console.
- It stays active until stopped, **idle for more than 10 days**, or unable to read its input.
- Objects beyond MaxConcurrentTaskCount queue in an **SQS queue created for you**.
- An object not sent to a worker within **14 days expires**.

> ON THE EXAM
>
> This is the pattern for continuously labelling production data, which is how you keep a training set current.

Ground Truth labels data **before** training.
A2I reviews predictions **after** deployment.

Human review of model predictions

- Routes **low-confidence predictions** to human reviewers automatically.
- Or routes a **random sample** of predictions, which gives you an ongoing audit.
- It removes the work of building a review system: task queues, UIs, reviewer management.
- Corrections come back as output data, and can feed retraining.
- Same three workforce options as Ground Truth: private, Mechanical Turk, vendor.

> ON THE EXAM
>
> The classic case is a document with poor handwriting: the model is unsure, a human decides, and the business keeps moving.

Turning corrections back into training data

- A2I writes reviewer output to **S3**, containing the original prediction and the human answer.
- Those pairs are **labelled examples of exactly the cases your model finds hard**.
- Feeding them into the next training run is the highest-value retraining data you have.
- They are also biased towards difficulty. Do not train on corrections alone.
- An EventBridge rule on human loop completion can trigger the downstream pipeline.

> ON THE EXAM
>
> Human review is usually sold as a safety net. Its second value is producing training data money cannot easily buy.

Three resources you must create

RESOURCE	WHAT IT DEFINES
`Worker task template`	What the reviewer sees: the data, instructions, and tools
`Flow definition`	The workflow: workforce, conditions, task config, output location
`Human loop`	One review task, created when conditions are met or by your code

ON THE EXAM

The flow definition is the one exam questions name. It is also called the human review workflow.

Built-in task types, and custom ones

	BEHAVIOUR
Built-in: Textract	Activation conditions available. Confidence, missing keys, sampling
Built-in: Rekognition	Activation conditions available. Confidence threshold on moderation
Custom: anything else	NO activation conditions. You call StartHumanLoop yourself
Consequence	Every StartHumanLoop call results in a human review. You do the filtering

ON THE EXAM

Textract and Rekognition are the only built-in task types. A custom SageMaker AI model is a custom task type.

Confidence threshold, or random sample

- **Confidence threshold:** only predictions below it go to humans. Catches the uncertain cases.
- **Random sampling:** a percentage of all predictions go to humans, confident or not.
- Sampling is how you **audit** a model, because it also checks confident predictions.
- A confidence threshold alone never questions a prediction the model is sure about.
- Thresholds are adjustable at any time, so you tune the accuracy against cost balance.

> ON THE EXAM
>
> A confidently wrong model is invisible to a confidence threshold. That is exactly what sampling exists to catch.

Ground Truth or Augmented AI?

GROUND TRUTH, WHEN

You need a dataset

- Data has no labels yet.
- The output is a training set.
- It runs before training, as a batch.
- Volume is known up front.

A2I, WHEN

You need oversight

- A model is already making predictions.
- The output is corrections and an audit trail.
- It runs continuously in production.
- Volume depends on model confidence.

THE TIEBREAKER

Before training or after deployment. That question separates them every time, and questions rarely make it harder than that.

What human review costs you

- A2I bills **per object reviewed**, so cost scales with how often humans are pulled in.
- A threshold set too high sends everything to humans and removes the point of the model.
- Mechanical Turk is priced per task and is much cheaper than a private workforce.
- A private workforce costs salaried time, which does not appear on the AWS bill at all.
- Automated data labelling reduces Ground Truth cost by removing human seconds.

> ON THE EXAM
>
> The confidence threshold is a cost dial as much as an accuracy dial. Tuning it is a legitimate exam answer.

Human feedback for foundation models

- Humans now label **model behaviour**, not just data. That is the shift.
- **Supervised fine-tuning data:** humans write ideal responses to prompts.
- **RLHF:** humans rank and compare model responses rather than labelling inputs.
- **Human evaluation** to choose between foundation models for a use case.
- Same machinery, different artefact: prompt-response pairs and preferences, not bounding boxes.

> ON THE EXAM
>
> This is where labelling meets the FM path. Module 18 covers human-in-the-loop evaluation properly.

What to reach for now, per AWS guidance

IF YOU NEED	CURRENT DIRECTION
Labelling for an existing account	Ground Truth still works. Nothing changes for you
Labelling in a new account	Not available. Vendors, partner tools, or your own workflow
Human review of Textract or Rekognition	A2I if you already have it. Otherwise build the loop yourself
Foundation model evaluation	Amazon Bedrock Evaluations, including human evaluation
Ground Truth Plus	Ended 30 June 2026. No successor of the same shape

ON THE EXAM

On the exam, answer as the guide expects. In an architecture review, say what is actually available.

Symptoms, and what they usually mean

SYMPTOM	USUAL CAUSE
Workers disagree constantly on the same objects	Ambiguous instructions, not careless workers
Labelling cost far above estimate	No automated data labelling, or too many workers per object
Human review queue growing without limit	Confidence threshold set too high
Confidently wrong predictions never reviewed	Threshold routing only. Add random sampling
Custom A2I workflow reviews everything	Custom task types have no activation conditions, by design
Streaming labelling job stopped by itself	Idle for more than 10 days, or input became unreachable

ON THE EXAM

The fifth row is the one to remember. It is documented behaviour, not a misconfiguration.

Spend human attention where the **model is uncertain**, and nowhere else.

A hospital group needs 30,000 chest X-ray images annotated with bounding boxes around suspected abnormalities. The images contain patient identifiers in the DICOM metadata.

Annotation requires radiological knowledge. The project has a fixed budget and the compliance team must approve the approach.

Which workforce should be used?

A Amazon Mechanical Turk, with instructions describing what an abnormality looks like

B Amazon Mechanical Turk for a first pass, with a private workforce verifying the results

C A public workforce, after stripping identifiers from the DICOM metadata

D A private workforce of the hospital group's own radiologists

◆ PAUSE AND DECIDE

Which workforce should be used?

A. Amazon Mechanical Turk, with instructions describing what an abnormality looks like

Fails twice. AWS recommends Mechanical Turk for non-sensitive data, and no instruction sheet turns a general worker into a radiologist.

B. Amazon Mechanical Turk for a first pass, with a private workforce verifying the results

Sounds efficient, and still exposes protected health information to a public workforce. Verification does not undo the exposure.

C. A public workforce, after stripping identifiers from the DICOM metadata

The strongest distractor. De-identification helps, but a chest X-ray is itself clinical data, and the expertise problem is untouched.

D. A private workforce of the hospital group's own radiologists

Two constraints, one answer. Sensitive clinical data stays inside the organisation, and radiologists supply the expertise the task requires.

THE SIGNAL

Two independent constraints were planted: sensitivity and expertise. Both point at private, so cost cannot outvote them.

A team deploys a custom SageMaker AI model that classifies insurance claims. They create an A2I flow definition for a custom task type and set an activation condition to route predictions below 70 percent confidence to reviewers.

In testing, every single prediction is sent for human review, regardless of confidence. The reviewers cannot keep up.

What explains this behaviour?

A The confidence threshold is inverted, so it is routing predictions above 70 percent

B Custom task types do not support activation conditions, so every StartHumanLoop call creates a review

C The flow definition is missing a worker task template, so A2I defaults to reviewing everything

D The model is not returning confidence scores, so A2I treats them all as zero

◆ PAUSE AND DECIDE

What explains this behaviour?

A. The confidence threshold is inverted, so it is routing predictions above 70 percent

Would send roughly the confident portion, not all of them. The symptom is every prediction, which no threshold produces.

B. Custom task types do not support activation conditions, so every StartHumanLoop call creates a review

Documented behaviour. Activation conditions exist only for Textract and Rekognition. The application must decide when to call StartHumanLoop.

C. The flow definition is missing a worker task template, so A2I defaults to reviewing everything

A template is required to create the flow definition at all. Without it there is no working workflow to over-trigger.

D. The model is not returning confidence scores, so A2I treats them all as zero

Plausible-sounding, and it presumes A2I is evaluating conditions for a custom type. It is not evaluating them at all.

THE SIGNAL

The fix is not in the flow definition. It is in the application: only call StartHumanLoop when your own code decides confidence is low.

A content platform uses Amazon Rekognition for image moderation with A2I. Predictions below 85 percent confidence go to human reviewers. That queue is working well.

Over six months, several images that Rekognition classified as safe with 96 percent confidence were later reported by users as clearly inappropriate. Nobody had reviewed them.

What should be added?

A Raise the confidence threshold from 85 percent to 98 percent

B Switch the workforce from Mechanical Turk to a private team

C Add a random sampling percentage so confident predictions are also reviewed

D Replace A2I with a Ground Truth labelling job over the same images

◆ PAUSE AND DECIDE

What should be added?

A. Raise the confidence threshold from 85 percent to 98 percent

Catches this specific 96 percent case and floods the queue with everything between 85 and 98. It also still misses anything above 98.

B. Switch the workforce from Mechanical Turk to a private team

Improves review quality for images that reach a reviewer. These images never reached one, so the workforce is irrelevant.

C. Add a random sampling percentage so confident predictions are also reviewed

Sampling is confidence-blind, so it inspects predictions the model was sure about. That is the only routing that can catch confident errors.

D. Replace A2I with a Ground Truth labelling job over the same images

Ground Truth builds datasets before training. It is not a production oversight mechanism, and it would not review live predictions.

THE SIGNAL

A confidently wrong model is invisible to a threshold. Sampling exists precisely because thresholds cannot see it.

A retailer must label 900,000 product photographs by category before training a classifier. The categories are visually distinctive and most photographs are unambiguous.

A quote from a labelling vendor for the full volume exceeds the project budget by a wide margin. The team asks how to reduce cost without accepting poor labels.

What should they do?

A Use automated data labelling, so humans label only the images the model finds uncertain

B Reduce the number of workers per image from three to one across the whole dataset

C Label a random 10 percent and train on that, discarding the rest

D Move the whole job to Mechanical Turk and accept the lower label quality

◆ PAUSE AND DECIDE

What should they do?

A Use automated data labelling, so humans label only the images the model finds uncertain

High volume plus mostly unambiguous images is exactly its case. Humans label a sample, the model labels the confident majority, and you set the threshold.

B Reduce the number of workers per image from three to one across the whole dataset

Cuts cost roughly threefold and removes annotation consolidation everywhere, including on the genuinely ambiguous images that need it most.

C Label a random 10 percent and train on that, discarding the rest

Cheap, and it throws away 810,000 images. Active learning uses that data; this simply abandons it.

D Move the whole job to Mechanical Turk and accept the lower label quality

The scenario explicitly rules out accepting poor labels, and label noise propagates into every prediction the classifier makes.

THE SIGNAL

Two facts make this work: very high volume, and mostly unambiguous images. Active learning needs both to pay off.

What to take from this module

- **Sensitivity picks the workforce.** Private for confidential or expert work, never Mechanical Turk.
- **Ground Truth before training. A2I after deployment.** That question separates them.
- **Custom A2I task types have no activation conditions.** Your code decides when to review.
- **Threshold catches uncertainty. Sampling catches confident errors.** They are not interchangeable.
- **Automated data labelling** is the biggest cost lever on a large labelling job.

> ON THE EXAM
>
> Module 08 closes Domain 1's classical half: data integrity, bias metrics and compliance.